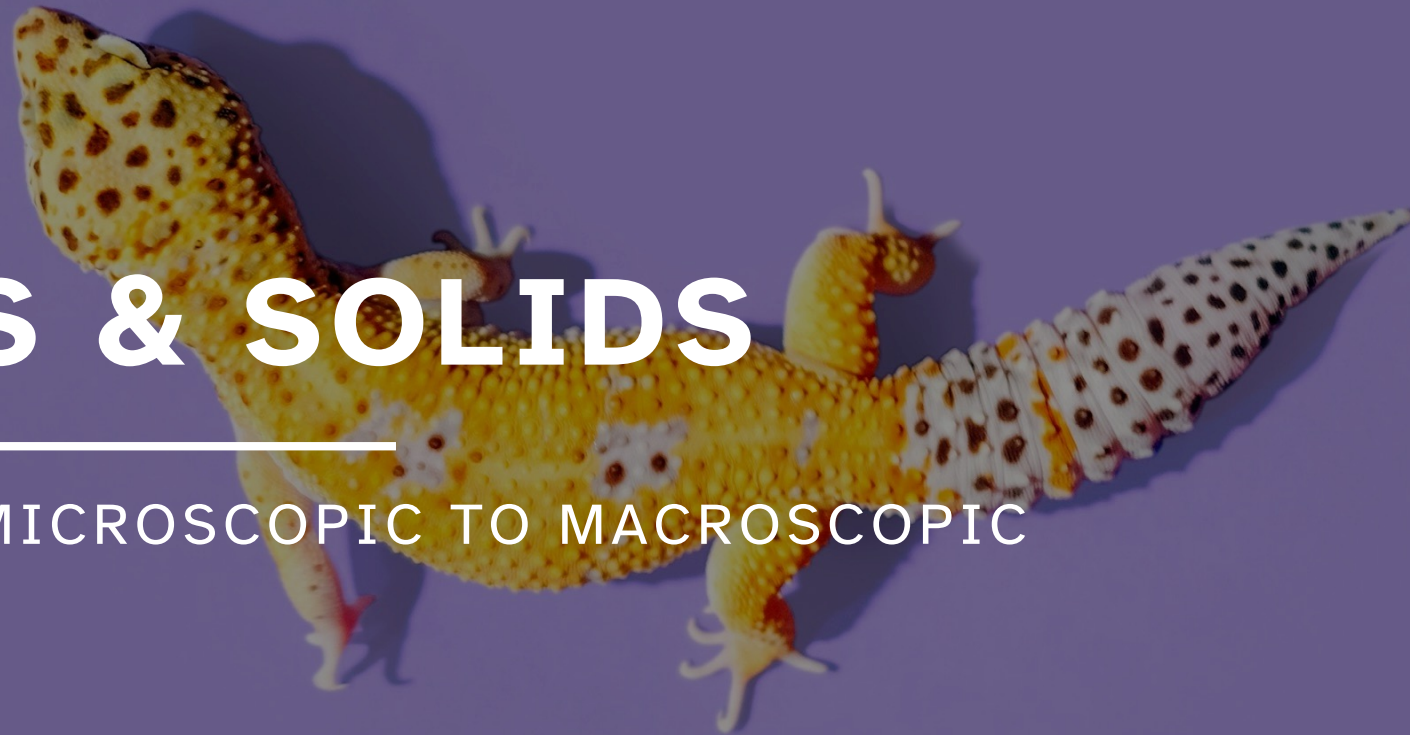


12A

LIQUIDS & SOLIDS

UNIT 04: FROM MICROSCOPIC TO MACROSCOPIC

CHEMISTRY 1310



LEARNING OBJECTIVES

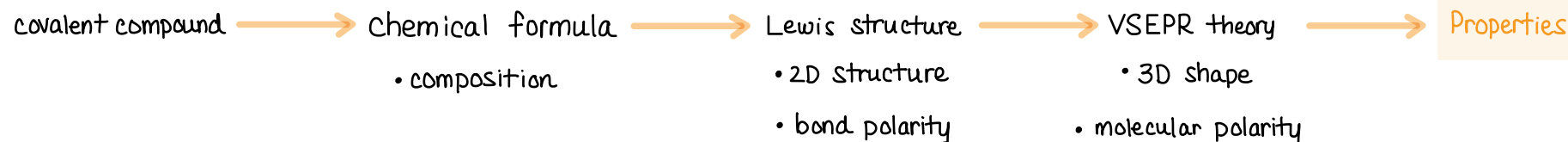
CHAPTER 12

1. List and describe the four types of intermolecular forces and identify the predominant intermolecular force in a substance.
2. Define and apply terms related to phase changes.
3. Calculate energy changes related to phase changes.
4. Interpret heating curves.
5. Predict physical properties (relative vapor pressures and relative phase change temperatures) based on strengths of intermolecular forces.
6. Define, describe, and interpret phase diagrams.
7. Describe the structures of various cubic unit cells.

Where did we start and where are we going?

Chemistry in Context

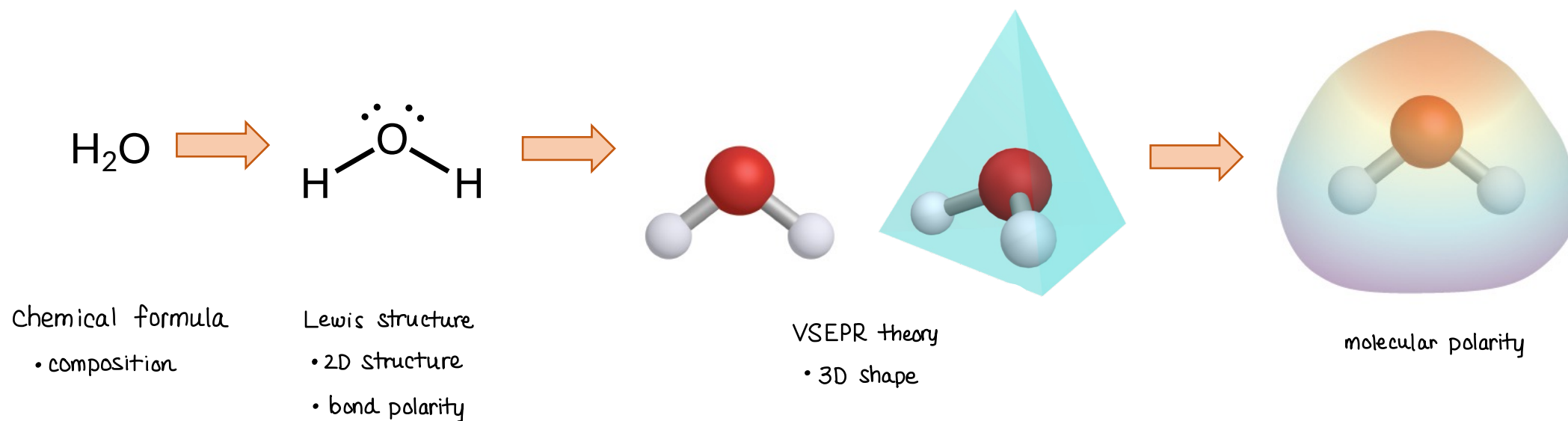
What is our overall goal?



Intermolecular Forces

12.1

Everything discussed so far has been regarding intramolecular forces, which are the chemical *bonds* that hold *atoms* together in a *single molecule*.



To think about the properties of *macroscopic or bulk samples*, we need to consider more than one molecule at a time.

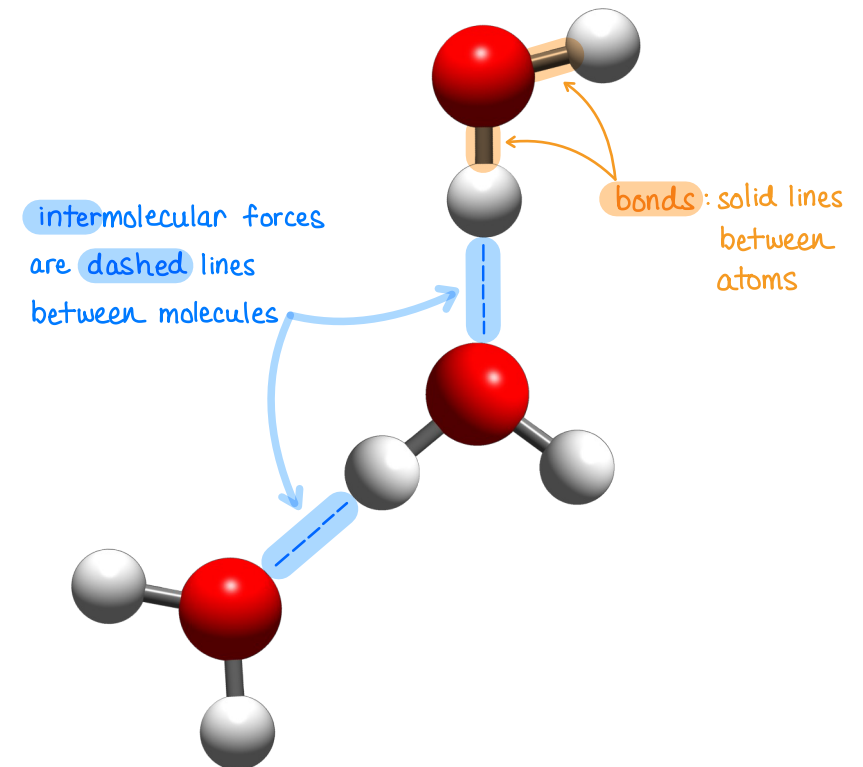
Intermolecular Forces

12.1

Intermolecular forces (IMFs) are the attractive forces that hold *individual molecules* together in a *sample*.

- IMFS are **MUCH** weaker than chemical bonds.
- Arise from *electrostatic* interactions between “charges” on adjacent molecules.

It is critical you differentiate intramolecular and intermolecular forces at a *microscopic* level!



Intermolecular Forces

12.1

Types of IMFS
depend mostly on
molecular polarity!

note: all IMFs are attractions

generally, strength increases →

	LDF	DD	HB	ID
	London Dispersion Forces	Dipole-Dipole Attractions	Hydrogen Bonding Interactions	Ion-Dipole Attractions
Nonpolar molecules & noble gases	dominant	×	×	×
Polar molecules	present but weak	dominant	×	×
Polar molecules that have H—O, H—N, or H—F bonds	present but weak	present but weak	dominant	×
Mixture of polar molecules + soluble ionic compounds	present but weak	maybe present but weak		dominant

Intermolecular Forces

12.1

London dispersion forces

LDF

- Present among all atoms and molecules.
- Generally, the weakest IMF.
- Strongest IMF when considering samples of *nonpolar molecules* or *noble gases*.
- Due to *temporary, instantaneous* dipoles: uneven e^- distribution

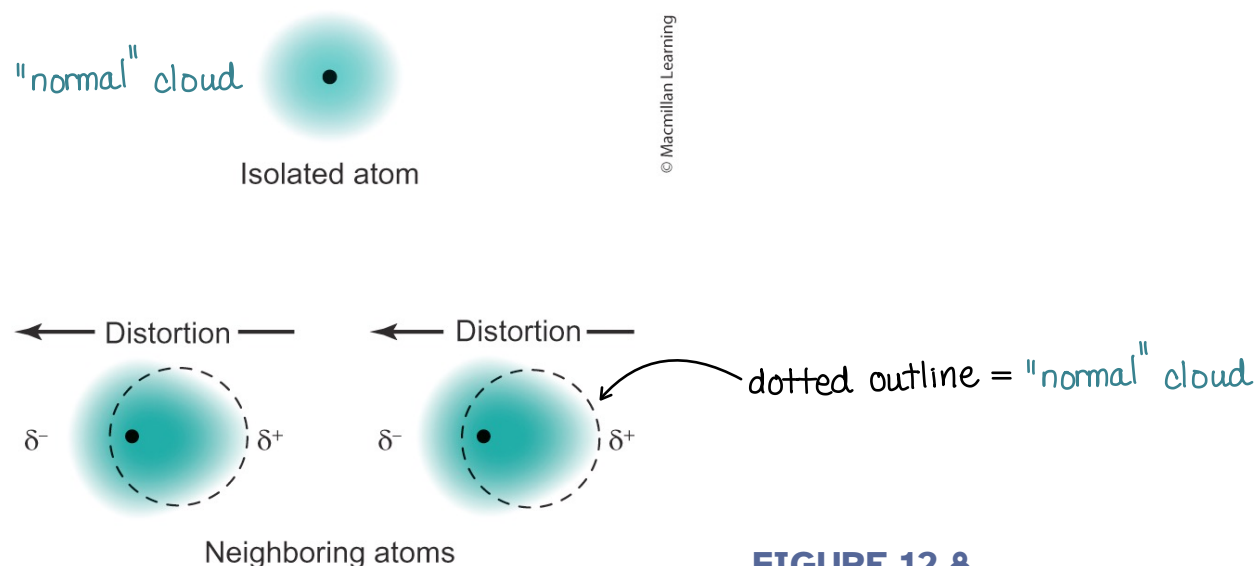


FIGURE 12.8

Intermolecular Forces

12.1

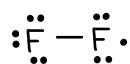
Strength of London dispersion forces depends on a molecule's polarizability.

- Ease with which the electrons (e^-) in an atom or molecule can be *distorted*.
- More polarizable $\rightarrow$ frequent and stronger temporary dipoles $\rightarrow$ *stronger* LDFs

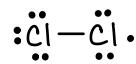
analogy: rock $\longrightarrow$ difficult to distort $\longrightarrow$ low polarizability
vs.
pillow $\longrightarrow$ easy to distort $\longrightarrow$ high polarizability

What makes a molecule *more polarizable*??

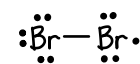
- Larger mass $\rightarrow$ more e^- $\rightarrow$ easier to distort
- Larger atoms/molecules $\rightarrow$ more e^- $\rightarrow$ easier to distort
- Longer, more “oblong” molecules $\rightarrow$ greater surface area $\rightarrow$ easier to distort



18 total e^-



34 total e^-



70 total e^-

Intermolecular Forces

12.1

Dipole-dipole attractions

DD

- Polar molecules have *permanent* molecular dipoles.
- The $\delta+$ of one molecular dipole is attracted to the permanent $\delta-$ of the molecular dipole on another molecule.
- *Common examples:* HCl, CH_2Cl_2 , any pure sample of a polar molecule



FIGURE 12.3

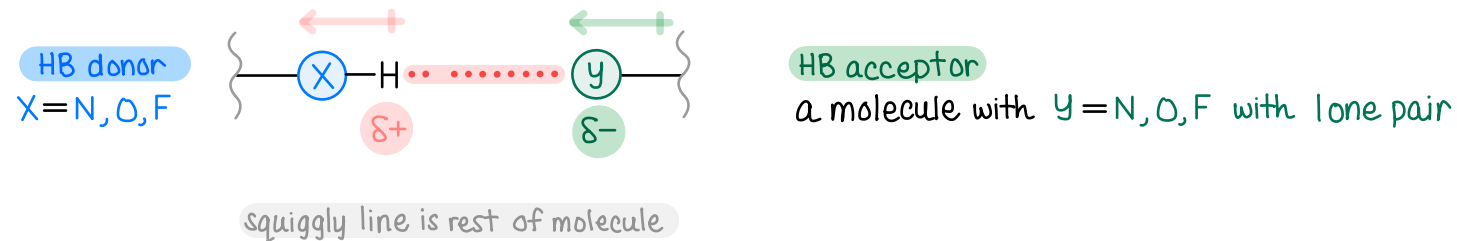
Intermolecular Forces

12.1

Hydrogen-bonding interactions

HB

- Special and *stronger* form of dipole–dipole attractions
- Strict Requirements:
 - HB donor molecule: polar molecule with a H atom *bonded* to N, O, or F
 - HB acceptor molecule: polar molecule with a lone pair on N, O, or F



H is very small and N, O, F are very small and electronegative (EN)

Very polar bonds (H—N, H—O, H—F have large ΔEN) → very strong molecular dipoles

Intermolecular Forces

12.1

Hydrogen-bonding interactions require us to think about the locations and orientations of lone pair electrons.

- (a) H_2O molecules
- (b) NH_3 molecules
- (c) HF molecules

FIGURE 12.5

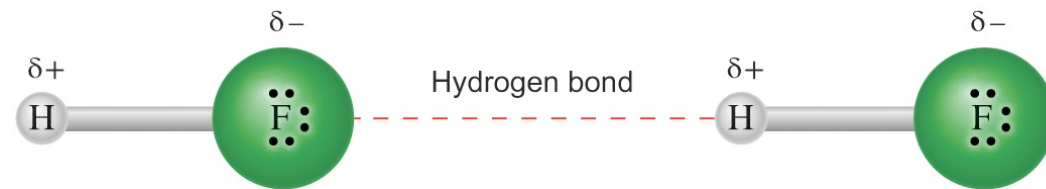
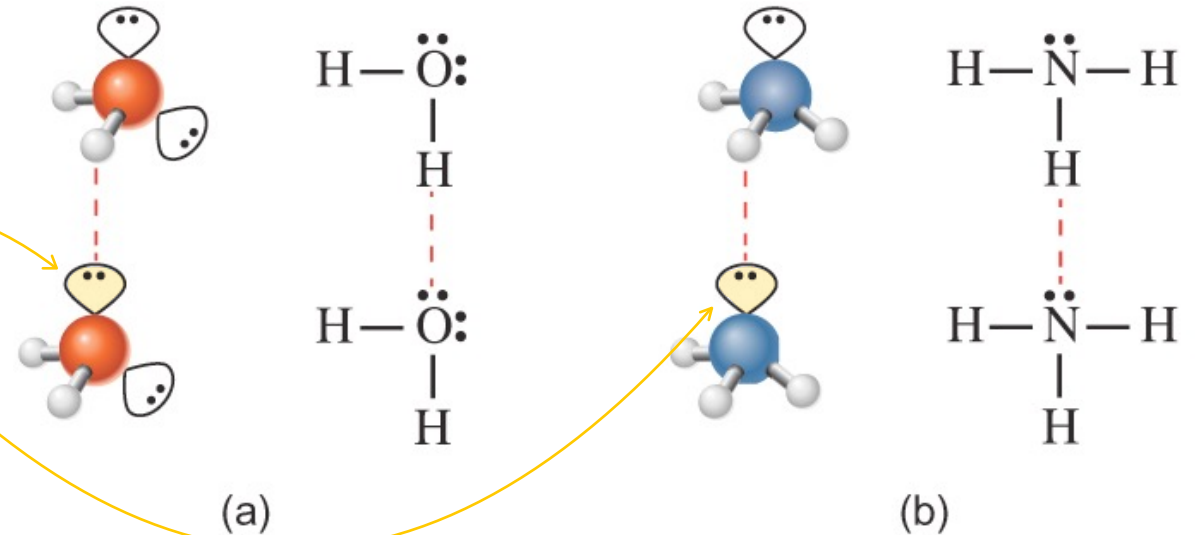


FIGURE 12.4

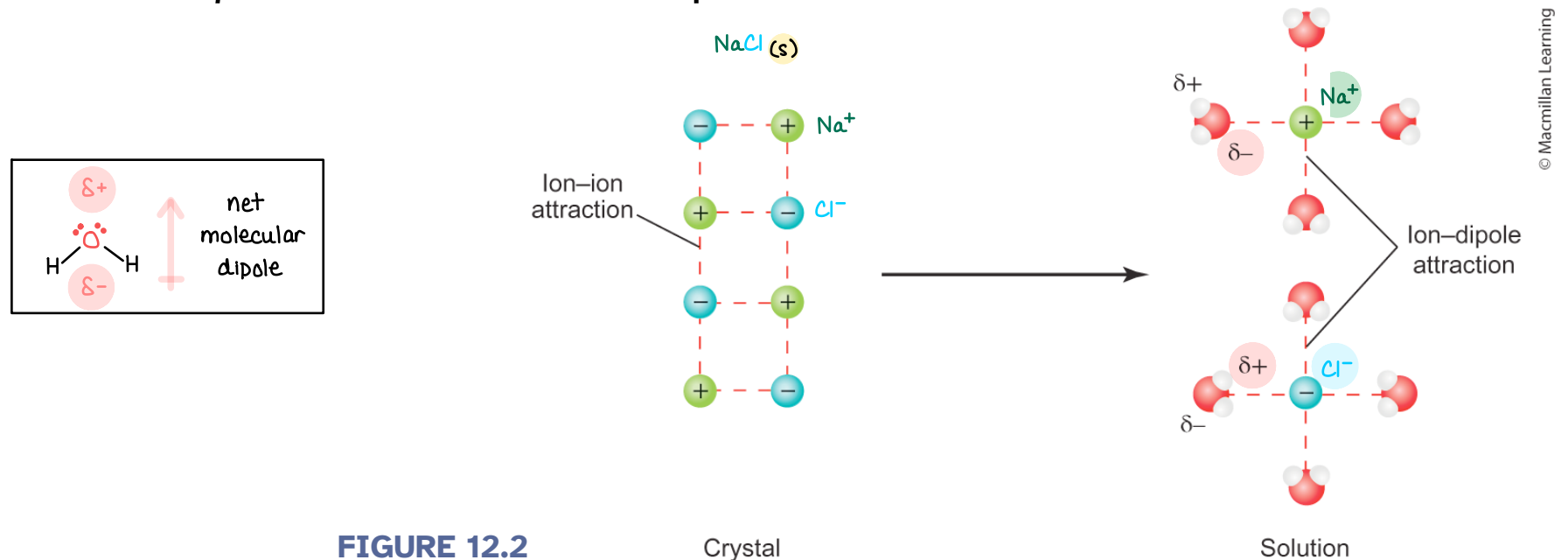
Intermolecular Forces

12.1

Ion-dipole attractions

ID

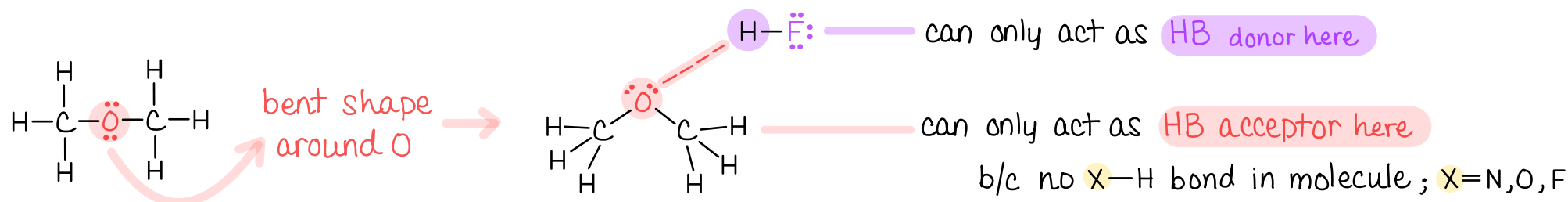
- Typically, the strongest of the IMFs.
- An ion (cation or anion) is attracted to the partial charge (of the opposite sign) of the permanent dipole in a polar molecule.
- *Most common examples:* soluble ionic compounds dissolved in water



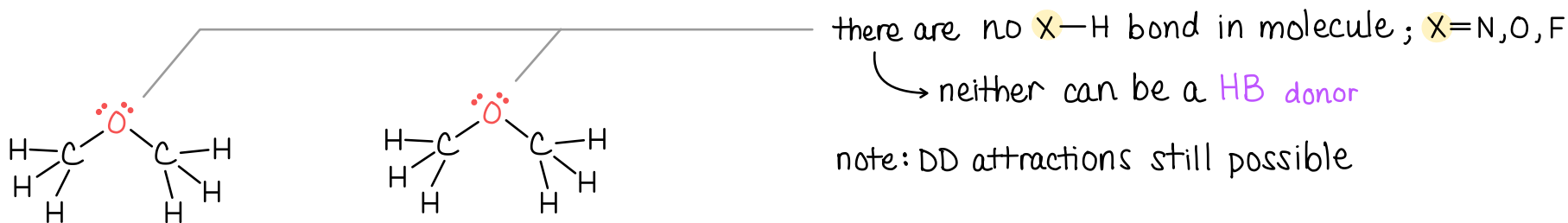
EXAMPLE PROBLEM

Learning Objective(s): 12.1

Can molecules of CH_3OCH_3 hydrogen bond to molecules of HF ? **yes!**



Can two molecules of CH_3OCH_3 hydrogen bond to one another? **no!**

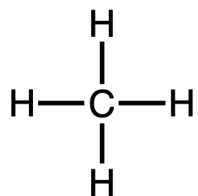


IN-CLASS QUESTION 1

Learning Objective(s): 12.1

i.e. strongest, dominant, most significant, etc.

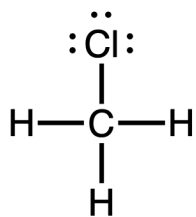
Identify the **primary/dominant** intermolecular force in pure samples of each of the following substances.



Substance A
CH₄

nonpolar

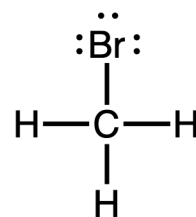
LDF



Substance B
CH₃Cl

polar

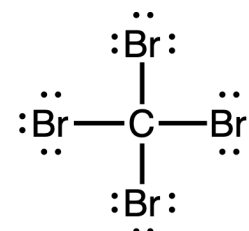
DD



Substance C
CH₃Br

polar

DD



Substance D
CBr₄

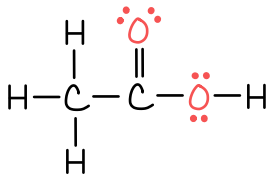
nonpolar

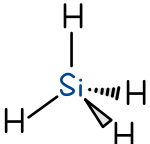
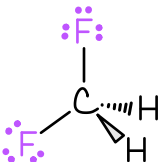
LDF

EXAMPLE PROBLEM

Learning Objective(s): 12.1

Determine the primary/dominant intermolecular force that is present in pure samples of each of the following compounds.

- CH_3COOH

polar
HB
- H_2S

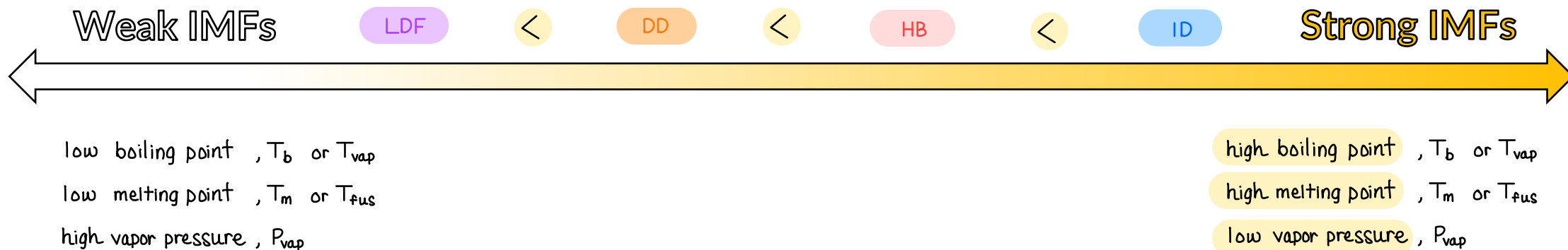
polar
DD
- SiH_4

nonpolar
LDF
- CH_2F_2

polar
DD

Intermolecular Forces

12.1

Physical molecular properties can be predicted from the types and strengths of IMFs.

Only IMFs are broken during phase changes (no chemical bonds are broken).



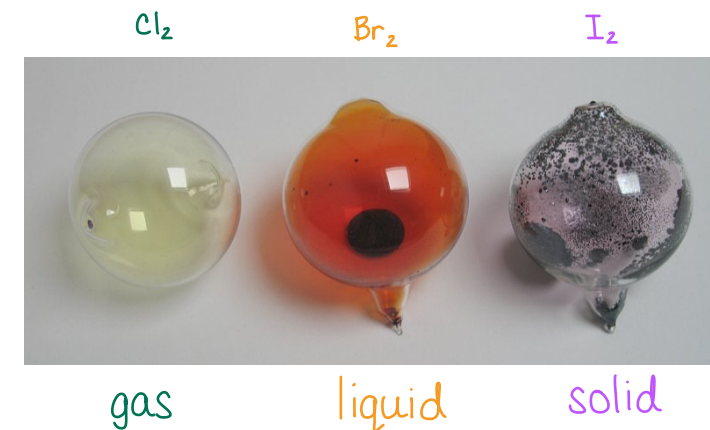
Intermolecular Forces

12.1

Relating polarizability, strength of LDFs, and melting/boiling points for the halogens.

all nonpolar

Molecule	Total Electrons	Molar Mass	Atomic Radius	Boiling Point	Melting Point
F ₂	18 total e ⁻	~38 g/mol	60 pm	85 K	53 K
Cl ₂	34 total e ⁻	~71 g/mol	100 pm	239 K	172 K
Br ₂	70 total e ⁻	~160 g/mol	117 pm	332 K	266 K
I ₂	106 total e ⁻	~254 g/mol	136 pm	457 K	387 K



increasing polarizability;
stronger LDFs

BONUS QUESTION

Assessed on: Fall 2021, Exam 3

Predict and arrange the boiling points of the following samples from lowest to highest.

